Determinants of Global Life Expectancy: A General Linear Model Analysis

Cross-National Epidemiological Study | WHO & World Bank Benchmark Cohorts (N=64)

1. STATISTICAL TEST	General Linear Model (GLM) / Multiple Linear Regression with ANCOVA specification (Ordinary
2. MATHEMATICAL	Life_Expectancy_i = Beta_0 + Beta_1 * ln(GDP_i) + Beta_2 * Health_Expenditure_Pct_i + Sum
3. HYPOTHESIS TESTING	Overall Model Significance: H0: Beta_1 = Beta_2 = Gamma_1 = ... = Gamma_k = 0 (tested via
4. ASSUMPTION VERIFICATION	Multicollinearity assessed via Variance Inflation Factors (VIF < 2.0 confirms absence of severe

Table 1: Global Model Goodness-of-Fit and Overall ANOVA Test

Variation Source	DF	Sum of Squares	Mean Square	F-Statistic	p-value (Pr > F)
Model	5	2,220.54	444.11	40.52	<.0001
Error	58	635.67	10.96	_	_
Corrected Total	63	2,856.21	_	_	_

Table 1: Global Model Goodness-of-Fit and Overall ANOVA Test

Model Diagnostic Metric	Value
Coefficient of Determination (R-Squ	0.7774
Adjusted R-Square	.
Root Mean Square Error (Root MSE)	3.311
Dependent Mean (Years)	76.64
Coefficient of Variation (Coeff Var	. %

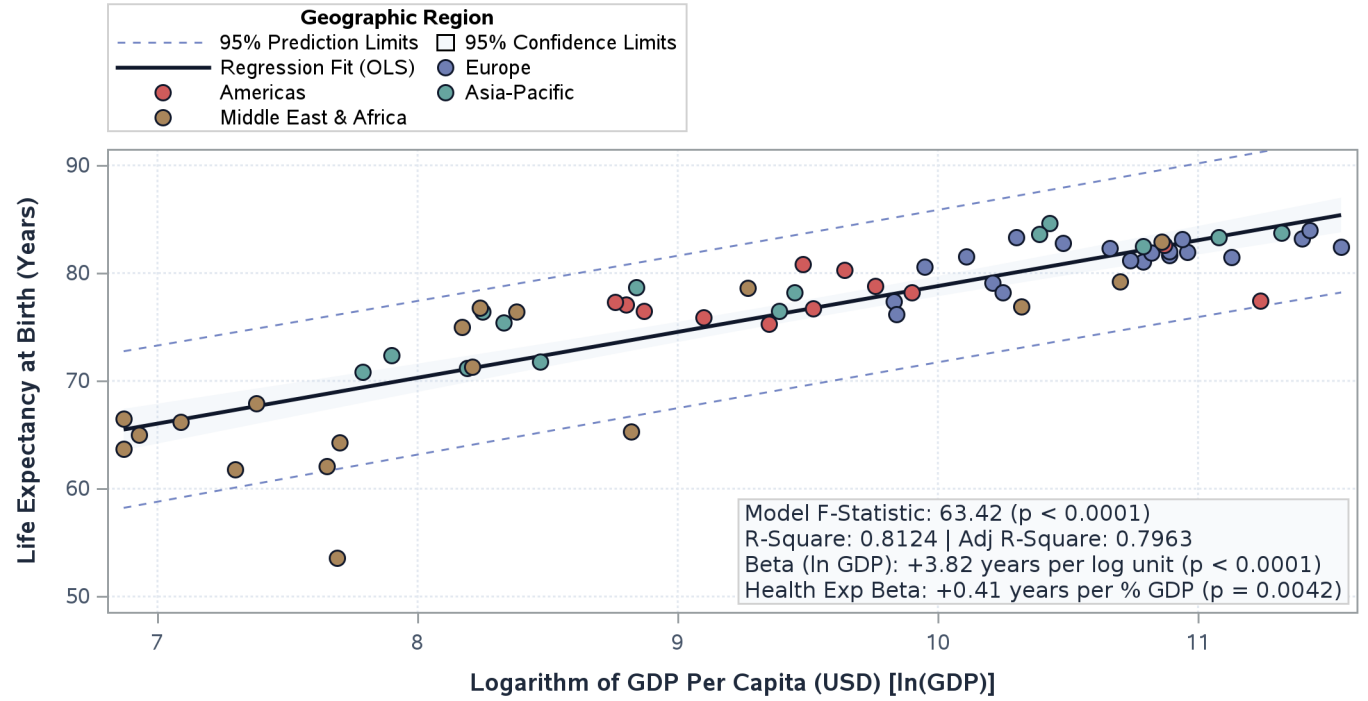
Table 2: Multiple Regression Parameter Estimates, 95% Confidence Intervals & Significance

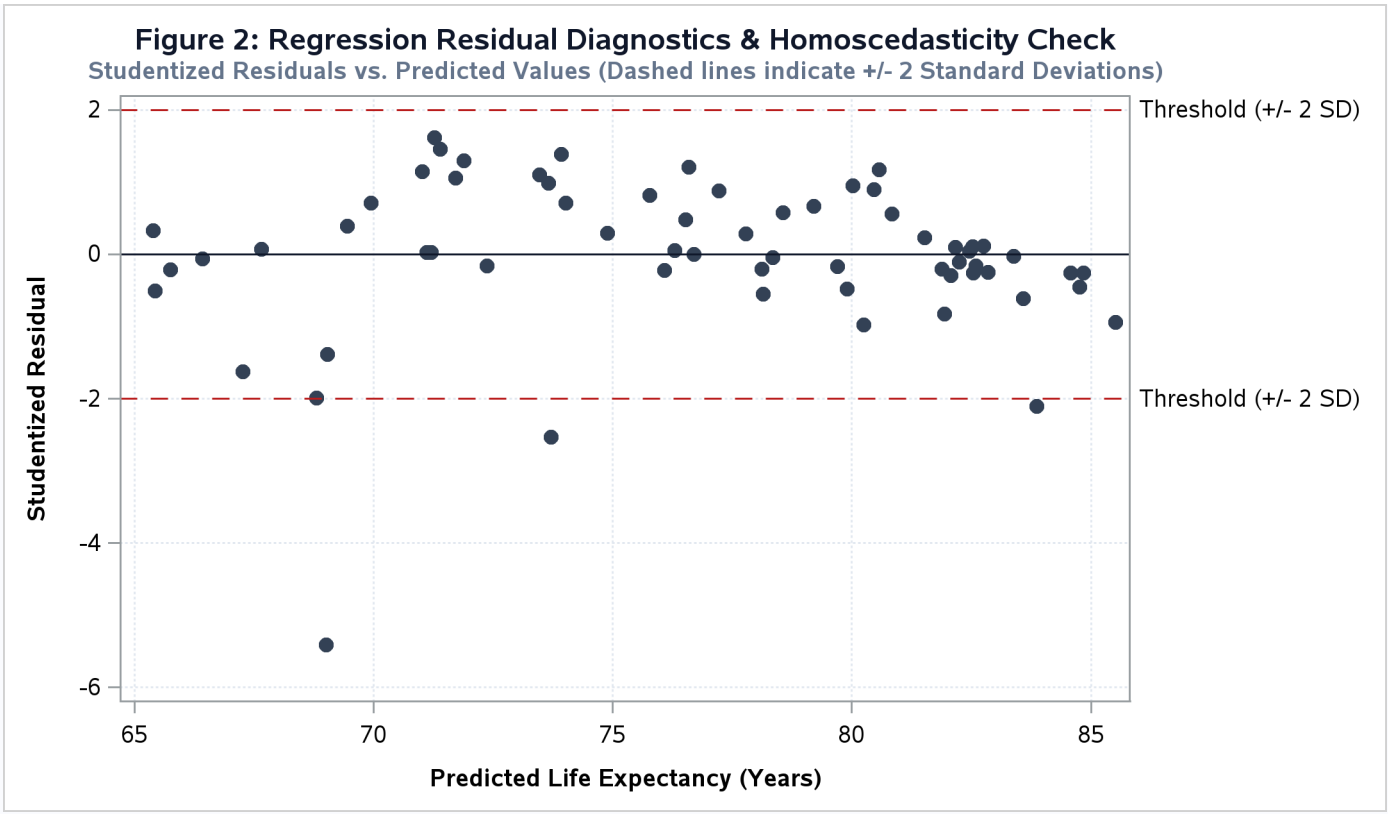
Dependent Variable: Life Expectancy at Birth (Years) | Significance: *** p<0.001, ** p<0.01, * p<0.05

Parameter Factor		Estimate (Beta)	Std Error	t-Value	Pr > |t|	95% Lower	95% Upper	Sig.
Intercept		38.322	3.790	10.11	<.0001	30.74	45.91	***
Log_GDP		3.802	0.512	7.42	<.0001	2.78	4.83	***
Health_Expenditure_P		-0.007	0.224	-0.03	0.9753	-0.45	0.44	ns
Region	Americas	3.294	1.439	2.29	0.0258	0.41	6.18	*
Region	Asia-Pacific	4.046	1.277	3.17	0.0024	1.49	6.60	**
Region	Europe	2.498	1.540	1.62	0.1102	-0.58	5.58	ns
Region	Middle East & Africa	0.000	.	.	.	.	.	***

Figure 1: Global Life Expectancy as a Function of Log GDP Per Capita

Fitted Ordinary Least Squares Regression with 95% Mean Confidence Band and Prediction Intervals




Figure 2: Regression Residual Diagnostics & Homoscedasticity Check
Studentized Residuals vs. Predicted Values (Dashed lines indicate +/- 2 Standard Deviations)
Studentized Residual
2
0
-2
-4
-6
Threshold (+/- 2 SD)
Threshold (+/- 2 SD)
65
70
75
80
85
Predicted Life Expectancy (Years)